3.2.2 About Coreless Motors

There are mainly two types of motors used in drones. One is the brushless AC motor, which is widely adopted in industrial drones for its high efficiency and reliability. The other is the ***coreless brushed DC motor (hollow cup motor)***, characterized by its lightweight design, compact size, and rapid response. It is commonly used in micro-drones, toy aircraft, and applications requiring high dynamic performance.

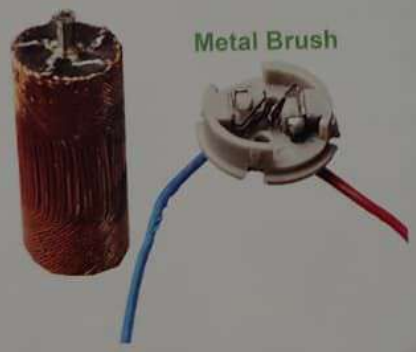
The coreless brushed DC motor (hollow cup motor) is primarily composed of components such as the ***rotor, stator, brushes, commutator, and drive circuit.***

Stator: The stationary part, typically made of permanent magnets arranged in a cylindrical structure, generating a fixed magnetic field.

Rotor: The rotating part, designed as a hollow cup-shaped coil (without an iron core). The absence of a traditional iron core reduces inertia and weight, enabling faster acceleration and deceleration.

Brushes and Commutator: The brushes (usually made of graphite or precious metal) maintain sliding contact with the commutator mounted on the rotor shaft. This mechanical system periodically reverses the current direction in the rotor windings to sustain continuous rotation.

Drive Circuit: A simple DC voltage control circuit (instead of a complex ESC) regulates the motor speed by adjusting the input voltage or pulse width modulation (PWM) signals.



Key Features of Hollow Cup Motors

High Power Density: Achieves high torque-to-weight ratios due to the coreless rotor design.

Low Electromagnetic Interference: Simpler brushed commutation generates less noise compared to brushless systems.

Rapid Response: Minimal rotor inertia allows millisecond-level torque response, ideal for precision control scenarios.

Low Cost: Simplified structure and absence of sensors reduce manufacturing complexity.

Moderate Lifespan: Brushes and commutators experience mechanical wear, limiting longevity compared to brushless motors (typically 200–500 hours).

The main parameters of a Coreless motor are as follows:

Motor Model:

The brushless motor specification for F8260 is 8260, where the numbers represent the diameter and height of the housing in millimeters, respectively. The larger the first two digits, the greater the motor diameter and torque; the larger the last two digits, the taller the motor and the better its dynamic balance. A motor that is both "fat" and "tall" has high torque and efficiency.

Motor KV Value:

The KV value refers to the increase in the motor's no-load RPM (revolutions per minute) for every 1V increase in voltage. For example, a KV value of 1000 means the motor spins at 1000 RPM at 1V, and at 10V, it spins at 10,000 RPM.

The KV value of a motor is related to the number of wire turns in its windings. A motor with more wire turns has a lower KV value, higher torque, and a lower maximum output current; a motor with fewer wire turns has a higher KV value, lower torque, and a higher maximum output current. At the same voltage, the lower the KV value, the slower the rotation speed and the higher the torque, allowing it to drive larger propellers; the higher the KV value, the faster the rotation speed and the lower the torque, making it suitable for smaller propellers.

3.2.3 About Propellers

Propellers are important components that directly provide lift for flight. When the propellers rotate at high speed driven by the motors, an air pressure difference is generated between the upper and lower layers of the propellers. The principle is the same as that of the wings of fixed-wing aircraft generating lift. When the thrust generated on the side with greater pressure is greater than or equal to the weight of the unmanned aerial vehicle (UAV), the UAV will take off.

Propeller Parameters:

- **Diameter:** It refers to the length of the propeller, and the unit is inches.
- **Pitch:** It refers to the distance that the propeller plane travels when it rotates one circle in the air, and the unit is also inches. This is related to the angle between the propeller surface and the horizontal plane. The larger the angle, the greater the pitch, and vice versa.

The F8620 drone uses one-piece propeller blades. Each propeller has a diameter of 66mm and a pitch of 45mm.

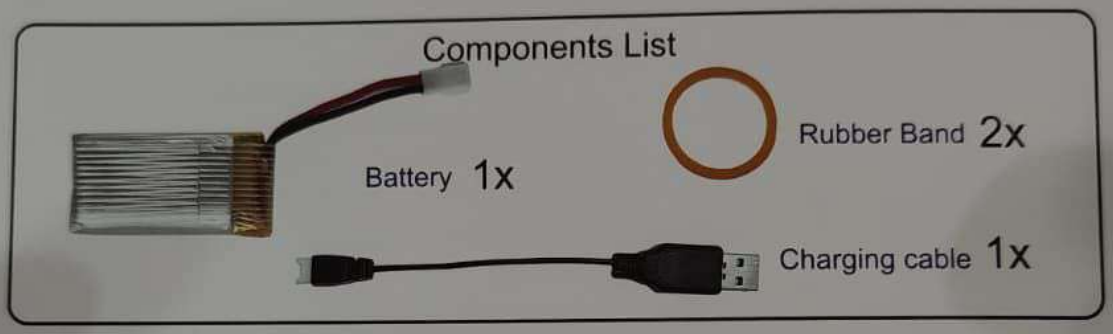
The propellers of UAVs are divided into clockwise and counterclockwise ones. Generally, the front side of the propeller is smooth and may have corresponding propeller parameter values at the same time. The clockwise propeller marked with "B" will generate pulling force when rotating counterclockwise, while the counterclockwise propeller marked with "A" will generate pulling force when rotating clockwise. Incorrect installation of propellers is a common mistake made by beginners. Incorrect installation will directly lead to a crash (the UAV being unable to fly or being damaged by falling).

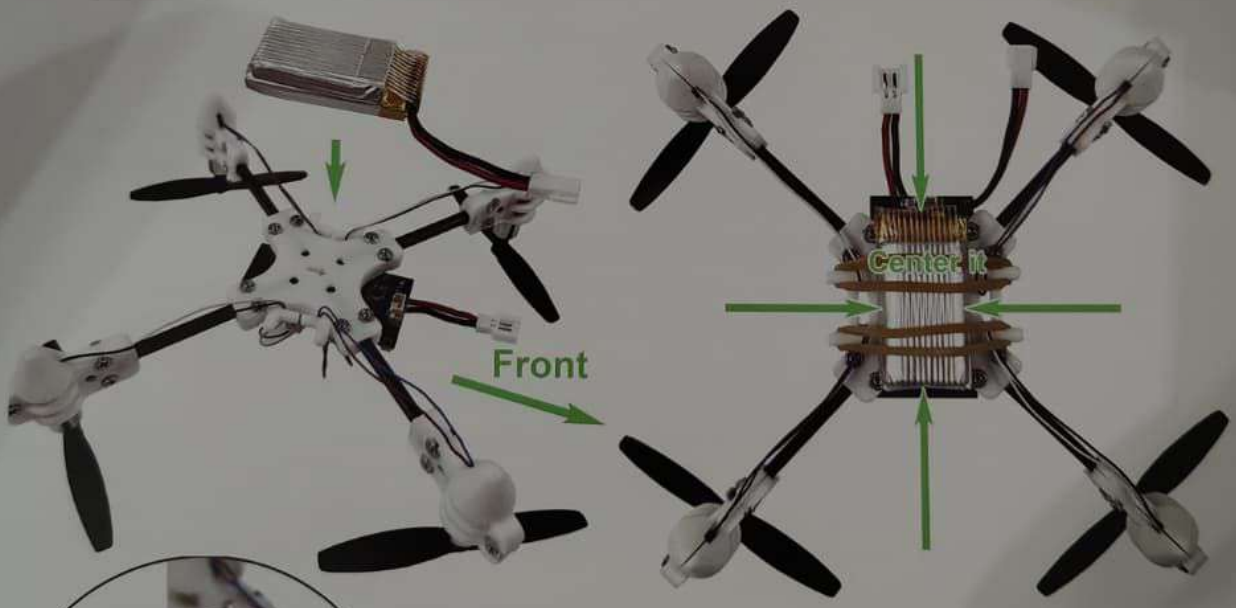
4 BATTERY

3.4.1 Assemble the Battery

Pocket 4

Assembled

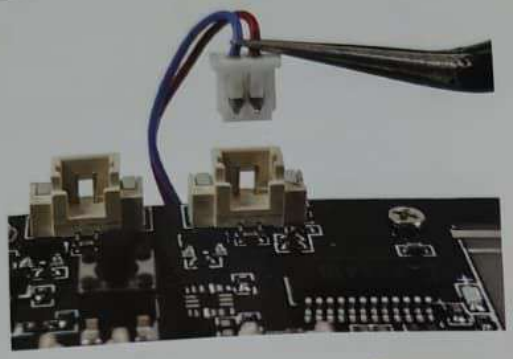


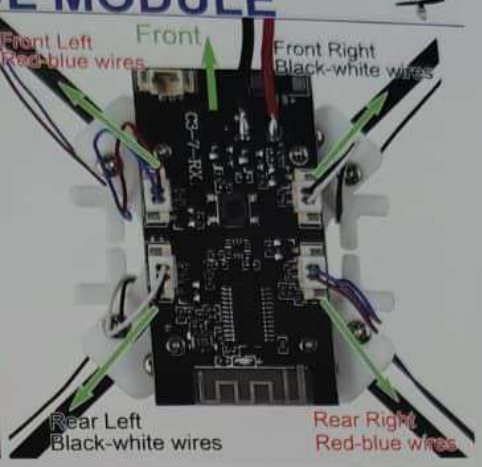


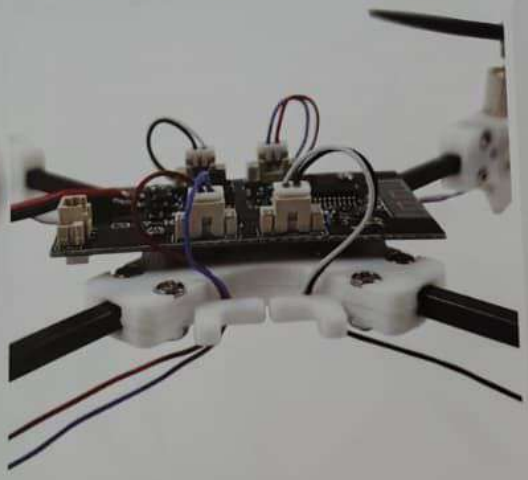
Tip:
When securing the battery, make sure it's centered both front-to-back and side-to-side. If not, it can throw off the drone's balance and cause unstable flight.
When plugging in the battery, check the terminals' direction to avoid damage.

3 FLIGHT CONTROL MODULE

2







Tip:
Plug the motor into the flight controller's matching connector as shown in the diagram. Check the connector's direction and handle gently to avoid damaging the pins.

Place the wire into the clip and gently straighten it to avoid leaving too much slack at the bottom.

3.3.2 Regarding the Flight Control Module

Flight Control Board (FCB), short for Flight Control Integrated Circuit Board, is a core component of unmanned aerial vehicles (UAVs), responsible for flight control and various complex functions.

Functions of the Flight Control Board:

a. Processing signals transmitted from the remote controller or automatic control signals of the UAV to achieve different functions based on the received signals.

b. Controlling the rotation speed of motors. The FCB sends signals to the electronic speed controllers (ESCs) to adjust the motor speeds, enabling the UAV to achieve stable hovering or movement.

c. Controlling external devices mounted on the UAV, such as gimbals, ultrasonic sensors, and servos.

3.3.3 Regarding Electronic Speed Controllers

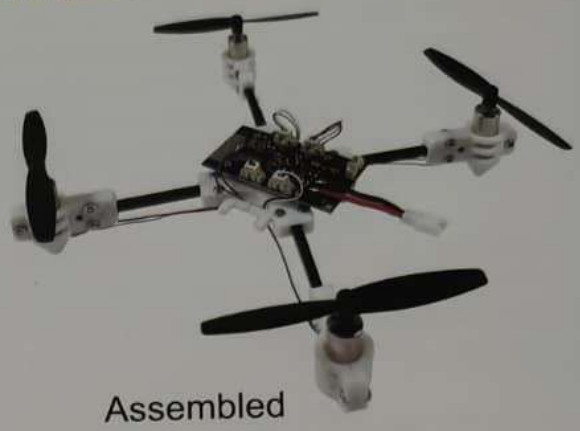
Electronic Speed Controller (ESC), often abbreviated as E-SC, is a general term for electric motor speed control systems. It serves as a crucial link between the motor and the flight control board. Its function is to receive signals from the flight control board and adjust the motor's speed accordingly, thereby influencing the flight attitude of the unmanned aerial vehicle (UAV).

The F8620 ESC, integrated onto the flight control board, reduces the complexity of the UAV.

3 FLIGHT CONTROL MODULE

3.3.1 Assemble the Flight Control Module

Pocket 3



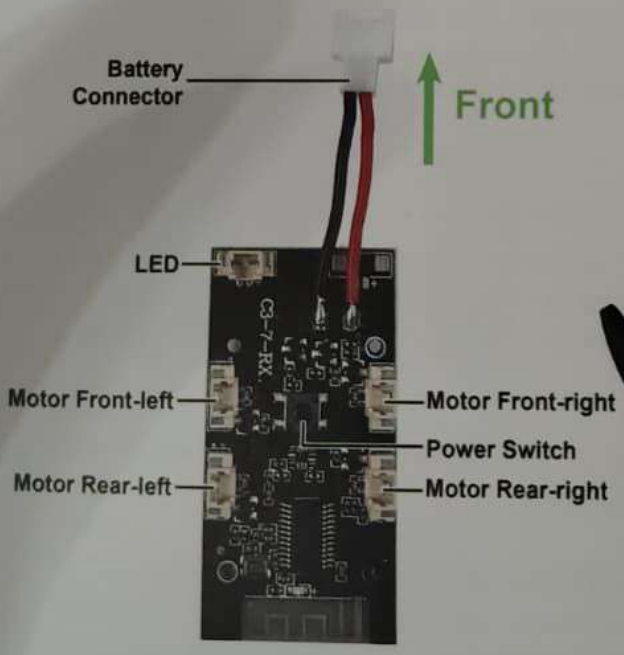
Assembled

Components List

Flight Controller 1x

Self-tapping screws M1.4*5 4x

1:1



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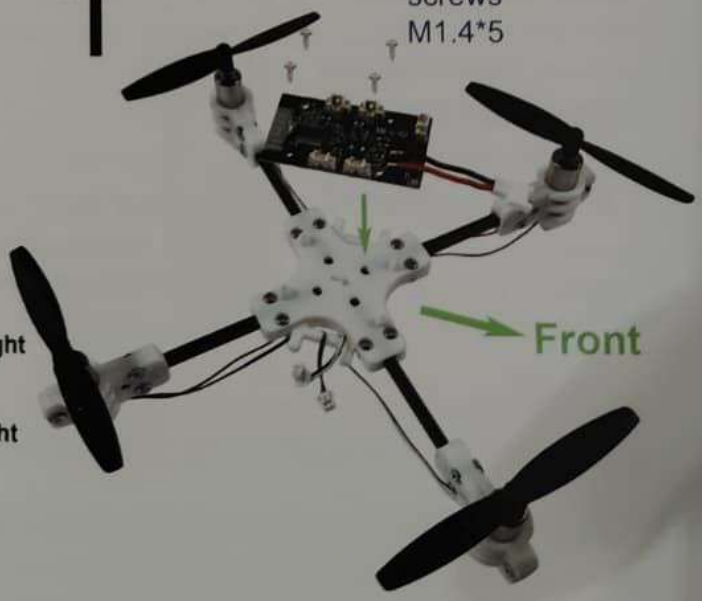

Tip:

The end of the flight controller with the power cable is the front.

Install it to match the arrow direction on the frame.

2 MOTOR BRACKET

The main parameters of a Coreless motor are as follows:

Motor Model:

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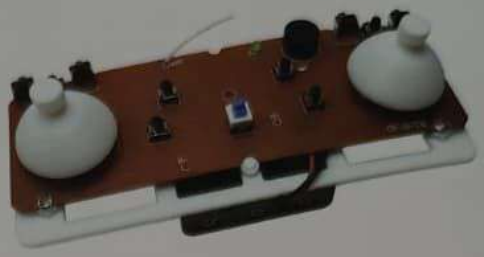
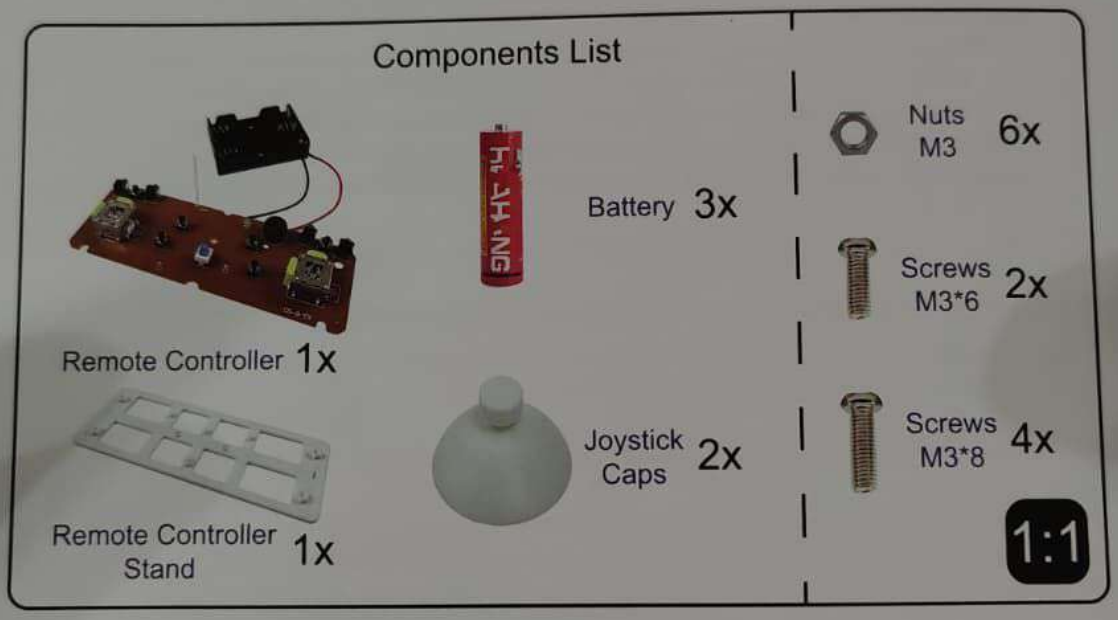
5 REMOTE CONTROLLER

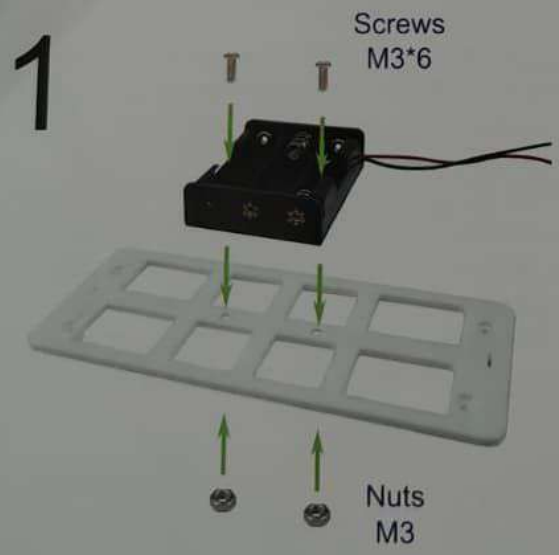
3.5.1 Install Remote Controller

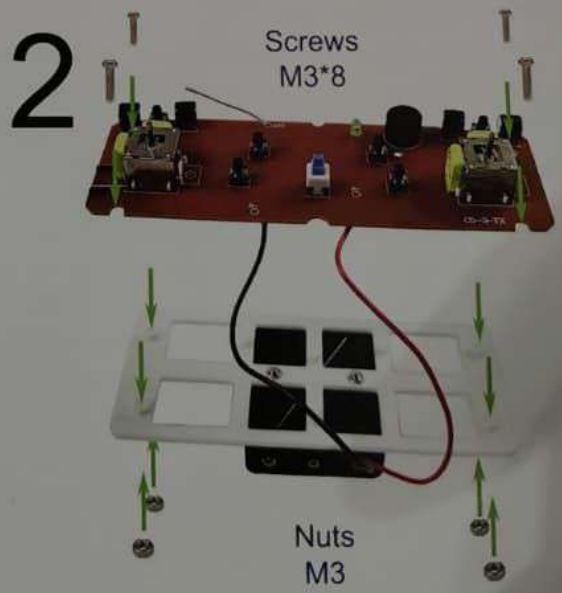
Pocket 5



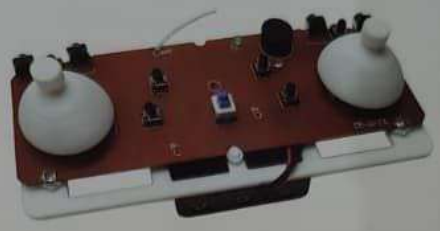
Assembled







5 REMOTE CONTROLLER



Tip:
The socket on the joystick cap and the joystick tab are both rectangular. Align them before inserting during installation.
When using it for the first time, we recommend placing the included button label card on the remote control.

3.5.2 Regarding the Remote Control

A remote control system consists of two parts: the transmitter and the receiver.

The remote control, also known as the transmitter, is a device used to remotely control machines, as its name suggests.

The corresponding component that receives signals is the receiver.

The function of the remote control is to send commands to the receiver on the drone, which then transmits the information to the flight controller for processing. When the flight controller receives commands from the ground, it executes the corresponding actions. Currently, the mainstream radio-controlled model aircraft remote controls on the market use the 2.4G signal format, which is characterized by stable signal transmission and long range.

Regarding the number of channels on a remote control, simply put, it refers to the number of control elements that the remote control can manage. The number of channels determines the functions that can be controlled on the aircraft. For a quadcopter drone, if you only need it to perform basic maneuvers such as ascending, descending, moving forward, backward, left, right, and rotating, then a 4-channel remote control is sufficient.

The receiver for the F8620 is integrated into the flight controller board.

Speed Setting

Trim Left

Headless Mode

Calibration

Trim Right

360° Flip

Up Throttle

Turn Left

Turn Right

Down Throttle

Automatic Takeoff

Automatic Landing

Power Switch

Trim Forward

Trim Backward

Move Forward

Move Left

Move Right

Move Backward

DO NOT ATTEMPT AT HIGH ALTITUDES to avoid drone loss, crash, and potential PERSONAL INJURY & PROPERTY DAMAGE.

EIELE

Operating Instructions

Power-on pre-flight procedures

Step 1
Insert the battery terminals into the power terminals on the flight control board, and briefly press the power button on the flight control board. The LED will start blinking slowly.

Step 2
Set the unmanned aerial vehicle on the open, obstacle-free and horizontal ground.

Step 3
Press the power button on the remote control. The LED on the remote control will remain lit, and the remote control and flight control board will begin automatic frequency matching.Once frequency matching is successful, the LED on the drone will remain constantly lit, indicating that the drone is ready.

Step 4
Briefly press the gyroscope calibration button on the remote control.The drone will start gyroscope calibration. The LED will start flashing rapidly. When the LED becomes constantly lit, the calibration is complete.(If the drone is tilted more than 30°, the LED will flash rapidly. Simply place the drone on a flat surface to restore the LED to a constantly lit state. This is a protective setting of the drone.)

Step 5
Gently push the left throttle stick up once, and the propellers will start spinning. (This is the pre-flight state. Observe if the drone is stable.)
Push the left throttle stick again to accelerate the propeller spin, and the drone will lift off the ground.

For the initial flight, hover the drone first to familiarize yourself with the basic flight rhythm of the drone and the functions of the buttons on the remote control. Try flying in different directions such as up, down, forward, backward, left, and right to get familiar with the control of the drone.

360° Flip

The LED on the drone remains lit when it is ready for operation, and it cannot be operated when the LED is blinking slowly.

1. Briefly press the 360° Flip button on the top right corner of the remote control, and the buzzer on the remote control will sound.
2. Flick the joystick on the right side once to the right. The drone will automatically perform a roll to the right.

P.S. Performing rolling maneuvers consumes a significant amount of battery power. By flicking the joystick in different directions in Step 2, the drone can roll forwards, backwards, left, or right.
This function should only be used in open spaces.
If the hovering effect is compromised after a roll, you can make fine adjustments or restart the drone.

Operating Instructions

Power-on pre-flight procedures

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This function should only be used in open spaces.
If the hovering effect is compromised after a roll, you can make fine adjustments or restart the drone.

3.4.2 Regarding Lithium Batteries

Batteries are vital components that provide electrical energy to the entire unmanned aerial vehicle (UAV) system, and lithium batteries are commonly used as the power source for UAVs. For multi-rotor UAVs, the energy payload per unit weight of the battery largely limits the endurance of the UAV. Simply increasing the battery capacity also increases its weight, failing to significantly enhance the UAV's endurance.

Common Battery Parameters

a. Voltage: The unit of voltage is denoted by V (Volt). The standard voltage of a single lithium battery cell is 3.7V, fully charged to 4.2V, and the protection voltage after discharging is generally set to 3.6V.

b. Capacity: The unit of battery capacity is milliampere-hour (mAh), which indicates how long a battery can continuously discharge at a certain current. For example, 2000mAh means the battery can continuously discharge at a current of 2000mAh (or 2A) for one hour.

c. Discharge Rate: The discharge capability of a battery is expressed in discharge rate (C), which indicates the maximum discharge current that can be achieved based on the battery's rated capacity. The formula for calculating the discharge rate (C) is: Discharge Rate = Discharge Current / Rated Capacity. For instance, a 2000mAh battery with a 25C discharge rate can achieve a maximum discharge current of 2A * 25 = 50A.

Estimation and calculation of battery usage time:

The motor on the F8620 operates at 1~3A, and the lithium battery capacity is 600mAh. Each motor's operating current is calculated at 1.5A.

*Flight time ≈ 600mAh× 60min / (1.5A*4pcs) = ~6 minutes (hover mode).*

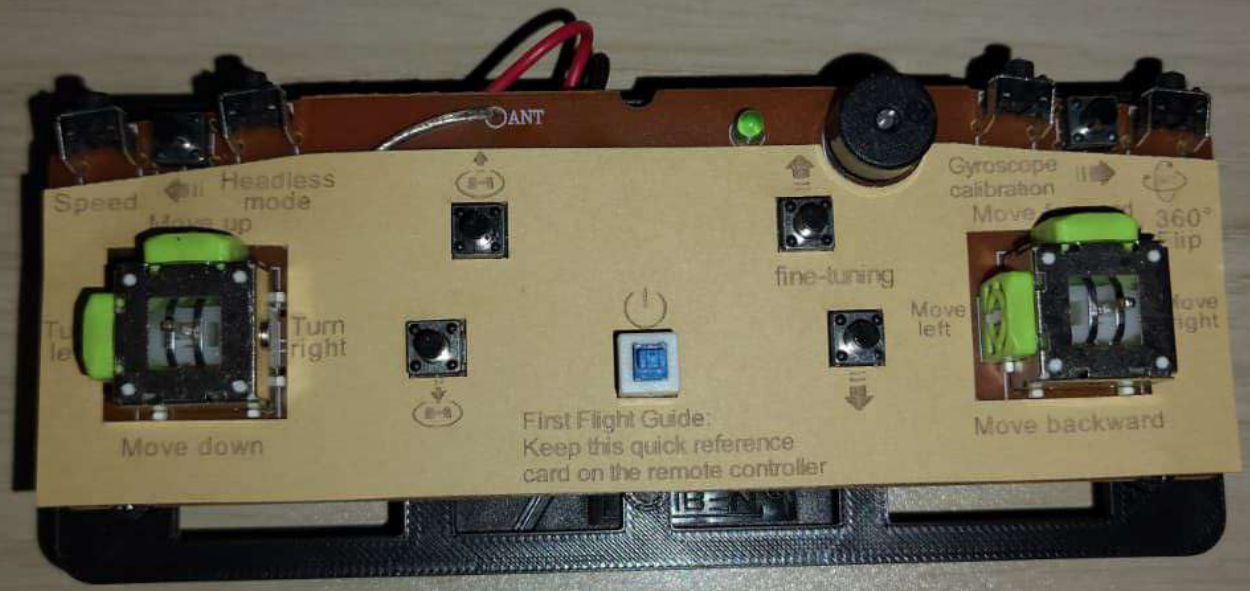
This calculation uses the motor current in hover mode. During continuous climbing, long-distance flight, or rolling maneuvers, the motor current will increase, shortening the flight time. There seems to be an error in this flight time calculation.

Battery protection:

When charging lithium batteries, a dedicated lithium battery balancer charger should be used. Lithium batteries that are not used for a long time should be discharged to around 3.85V for storage. (This is also why you were instructed to charge the battery at the beginning of assembling the kit.)

When using lithium batteries, it is crucial to avoid overcharging or over-discharging, as this can cause irreparable damage to the battery, such as swelling or damage to the battery cells. The lithium battery of the F8620 is equipped with high-voltage and low-voltage protection boards, and low-voltage protection for the battery is also programmed into the flight control system. Charging will stop if the voltage exceeds the protected 4.2V, and flight will cease if the voltage drops below the set 3.6V during operation.

PS: Lithium batteries should not be squeezed, punctured, or heated to high temperatures. If the battery has already swelled, it is recommended to stop using it, dispose of it properly, and replace it with a new battery in a timely manner.


ANT
Speed
Headless mode
Gyroscope calibration
360°
fine-tuning
Move left
Turn right
Move down
Move backward
First Flight Guide:
Keep this quick reference card on the remote controller

ANT
B-
B-
C5-2-TX

ANT
B+
B−
C5-2-TX

1 FRAME

The F8620 quadcopter frame comprises three main components: a fuselage (3D-printed with PLA-Basic material), carbon fiber rods, and motor mounts (also 3D-printed with PLA-Basic material).

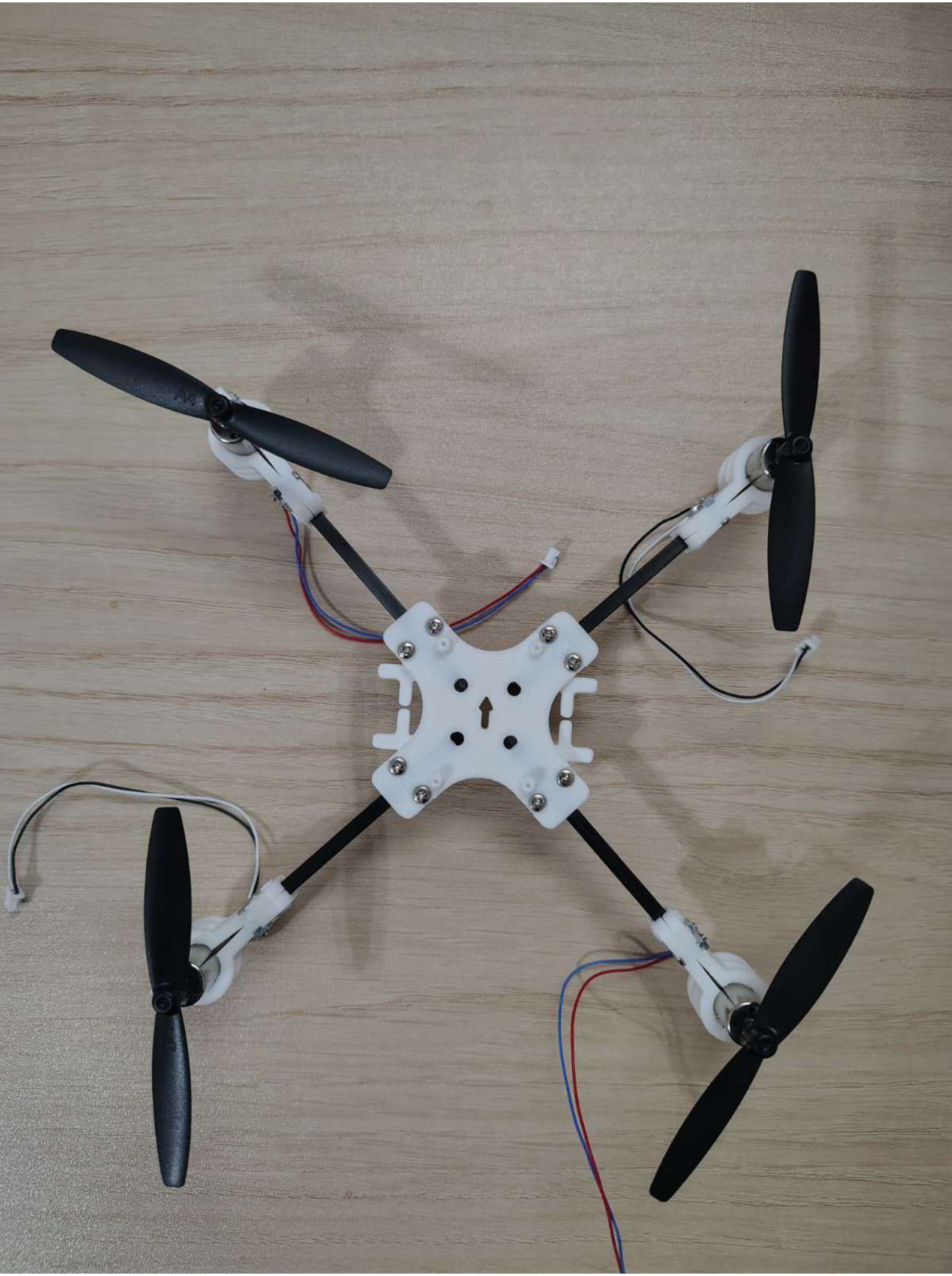
Of course, a thorough understanding of the drone, careful reading of the installation instructions, and high-quality assembly are decisive factors for the stable flight of the F8620, as even the best design cannot compensate for mistakes caused by incorrect installation.

3.1.3 About the Frame Structure

Frame Wheelbase:

The wheelbase of a drone refers to the distance between the central axes of two motors diagonally opposite, usually measured in millimeters. A larger wheelbase indicates a increases the drone's overall dimensions of the drone and enables the use of larger propellers.

The F8620 features a 145 mm motor-to-motor diagonal distance, with propellers of 66 mm diameter .

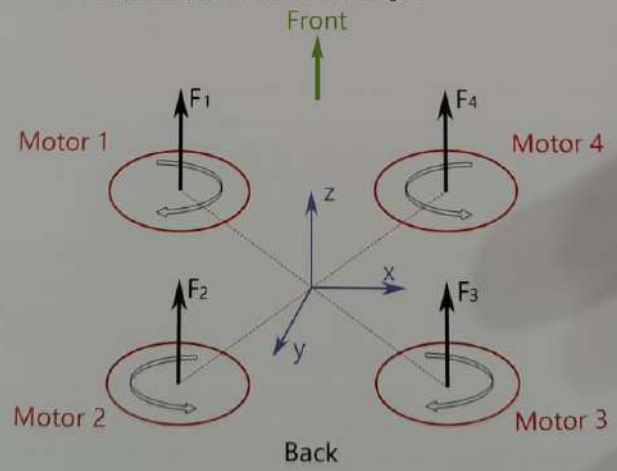


2. The Flying Principle

A simple quadcopter drone mainly consists of six parts: frame, motors, flight controller board, propellers, battery, and remote controller.

2.1 The Principle of Flight Attitude Control

The rotor structure of a quadcopter is shown in the figure.



2. The Flying Principle

In a quadcopter drone, the rotation directions of adjacent rotors are opposite. The rotors M1 and M3 rotate clockwise, while the rotors M2 and M4 rotate counterclockwise. During flight, the clockwise reaction forces (counter-torques) generated by M2 and M4 cancel out the counterclockwise reaction forces (counter-torques) generated by M1 and M3, allowing the drone's body to remain stable.

By ensuring that each rotor of the quadcopter drone rotates at the same speed (resulting in zero net torque), various movements can be achieved through adjustments to the rotation speeds of the four rotors.

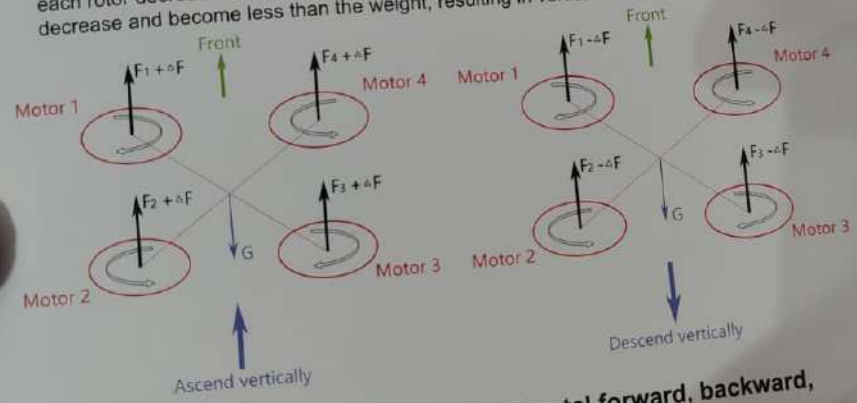
A quadcopter has a total of six degrees of freedom in space (including translational and rotational movements along the three coordinate axes). Control over each degree of freedom can be achieved by adjusting the rotation speeds of different motors.

Below is a detailed explanation of various flight attitudes of the aircraft.

a. Vertical movement (vertical ascent, vertical descent)

Ignoring external factors, when the total lift force F of the drone is equal to its weight, the drone will hover.

If each rotor increases its rotational speed by the same amount, the total lift force F will increase and become greater than the weight, resulting in vertical ascent. Conversely, if each rotor decreases its rotational speed by the same amount, the total lift force F will decrease and become less than the weight, resulting in vertical descent.



b. Pitch movement, roll movement (horizontal forward, backward, leftward, or rightward flight).

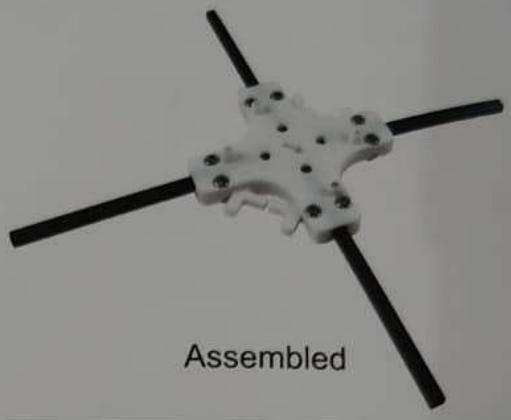
Disregarding external factors, when the rotational speeds of the front rotors M1 and M4 decrease while those of the rear rotors M2 and M3 increase, the lift generated by the rear rotors M2 and M3 exceeds that of the front rotors M1 and M4, causing the quadrotor to move forward. Conversely, if the rotational speeds of the front rotors increase while those of the rear rotors decrease, the quadrotor will move backward. This is also known as pitch movement.

1 FRAME

Start Assembling

3.1.1 Assemble the Frame

Pocket 1



Assembled

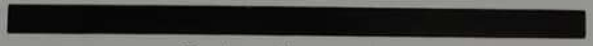
Components List

Upper frame 1x

Lower frame 1x

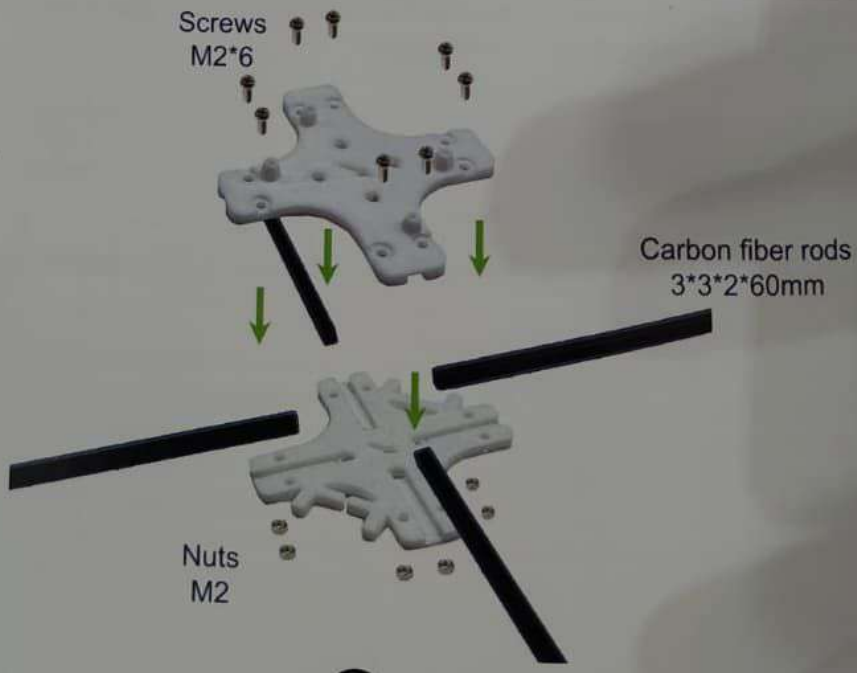
Screws M2*6 8x

Nuts M2 8x

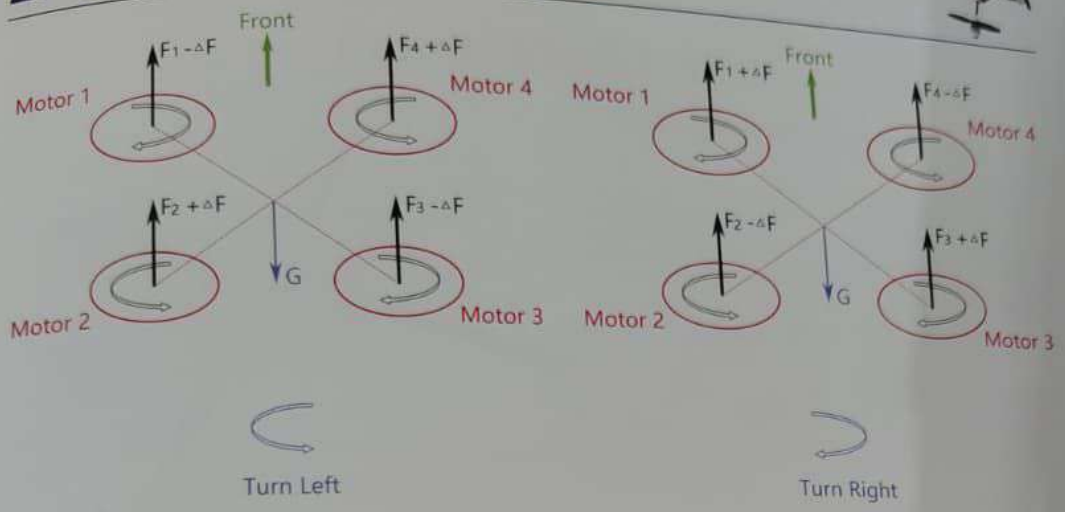


Carbon fiber rods 3*3*2*60mm 4x

1:1



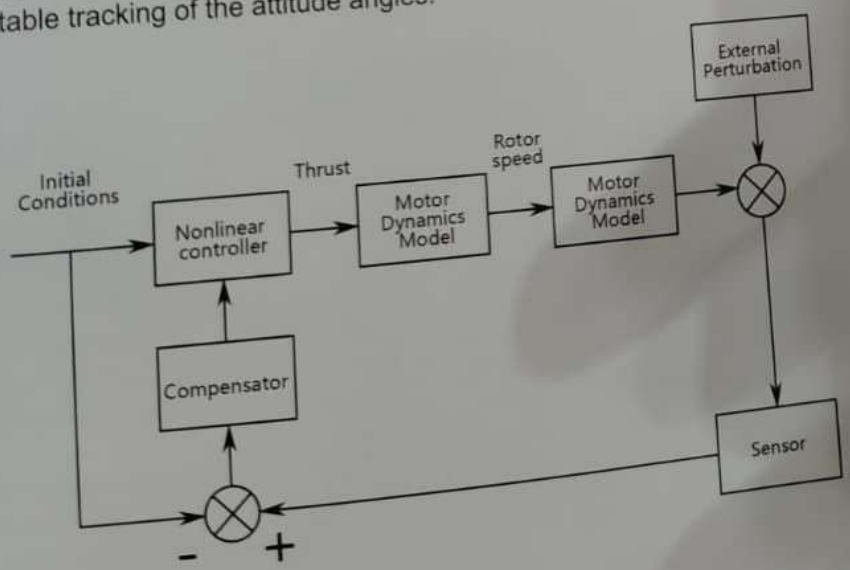
2. The Flying Principle



2.2 Signal Control Logic for Flight Attitude

In the autonomous control system of a quadcopter drone, attitude stabilization control is the foundation for achieving autonomous flight. Its task is to control the three attitude angles (pitch, roll, and yaw) of the quadcopter to stably track the desired attitude signals and ensure that the closed-loop attitude system exhibits the desired dynamic characteristics.

Due to the coupling characteristics between the attitude and translation of the quadcopter, it can be analyzed that only by ensuring stable attitude control can the total lift force of the rotors generate components in the desired direction, thereby controlling the aircraft to fly along the desired trajectory. Since there are uncertainties in the channels, such as model errors, environmental disturbances, observation errors, etc., these uncertainties will degrade the closed-loop performance of the system. Therefore, when designing the drone control system, it is necessary to consider the system's anti-disturbance performance, i.e., the robustness of the closed-loop system. As a result, a certain disturbance compensator needs to be designed to approximate and compensate for the disturbances, in order to achieve stable tracking of the attitude angles.



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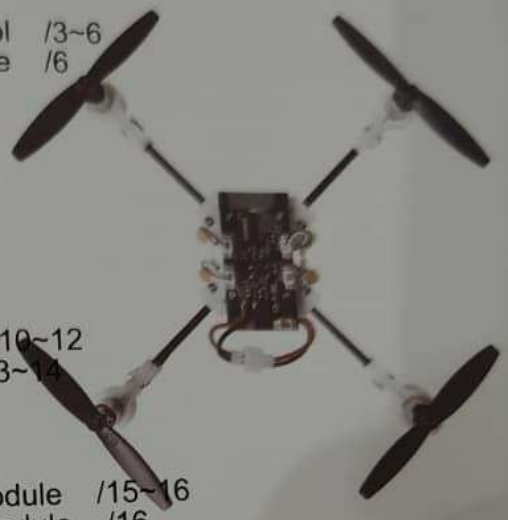
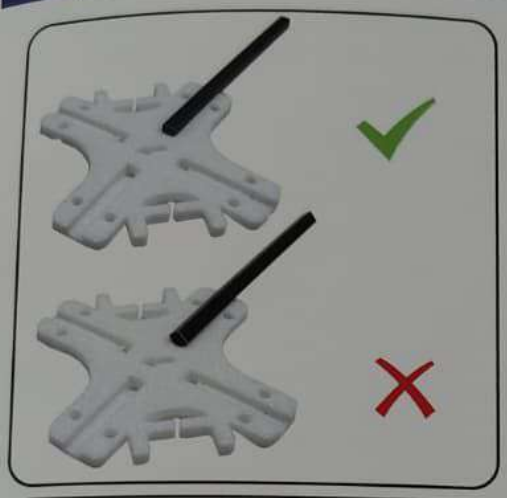


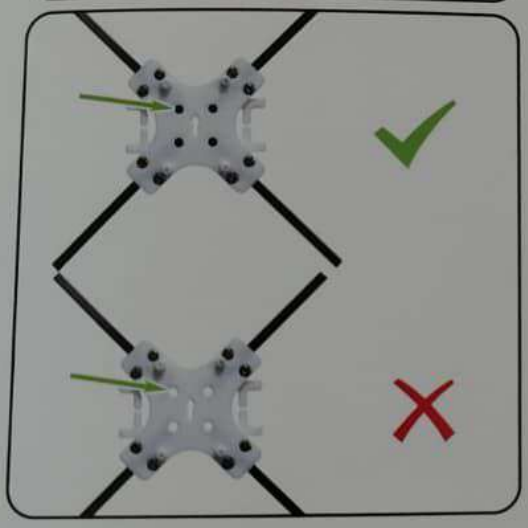
Remote Control Function Labels

The white 3D-printed parts in the installation guide images are for reference only. The actual product color may vary, and color differences only apply to the 3D-printed parts.

1 FRAME



Tip:
Initially, do not tighten the screws directly. After confirming that the carbon fiber pole is correctly installed, gradually tighten each screw one by one until the carbon fiber pole cannot be easily pulled out. Over-tightening the screws may cause the frame to deform, which could affect subsequent installations. (If white stripes appear on the frame, stop tightening the screws and loosen them slightly.)



Tip:
Check through the pre-drilled hole to see if the carbon fiber rod is properly installed. The rod is correctly in place when the black carbon fiber rod fills the hole completely.

3.1.2 About the Frame

As the skeleton part of the quadrotor drone, the frame plays an important role in supporting the entire drone and providing installation positions for each component, and is one of the indispensable components of the drone.

The frame's weight determines the drone's base weight, indirectly affecting its flight duration and payload capacity. The frame's sturdiness directly influences whether the drone's flight attitude can meet expectations. Excessive hollow spaces can cause resonance throughout the aircraft, directly affecting the operation of electronic gyroscopes and potentially leading to loss of control. An overweight frame may reduce propeller efficiency. Therefore, an optimal frame requires iterative design and testing.

The F8620 frame design has carefully considered these factors. It utilizes lightweight yet robust carbon fiber material and employs 3D printing technology for hollow-design features and manufacturing fixtures. These refined designs reduce weight while maintaining structural integrity.

1 FRAME

The F8620 quadcopter frame comprises three main components: a fuselage (3D-printed with PLA-Basic material), carbon fiber rods, and motor mounts (also 3D-printed with PLA-Basic material).

Of course, a thorough understanding of the drone, careful reading of the installation instructions, and high-quality assembly are decisive factors for the stable flight of the F8620, as even the best design cannot compensate for mistakes caused by incorrect installation.

3.1.3 About the Frame Structure

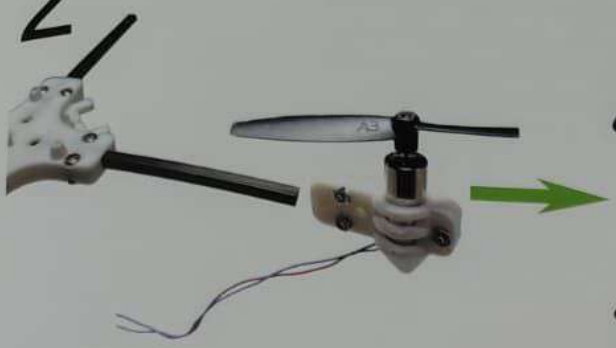
Frame Wheelbase:

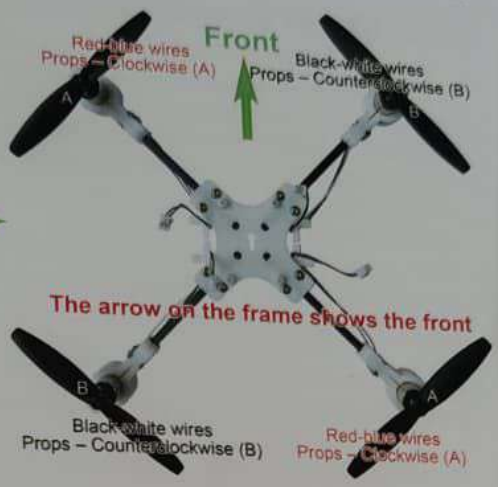
The wheelbase of a drone refers to the distance between the central axes of two motors diagonally opposite, usually measured in millimeters. A larger wheelbase indicates a increases the drone's overall dimensions of the drone and enables the use of larger propellers.

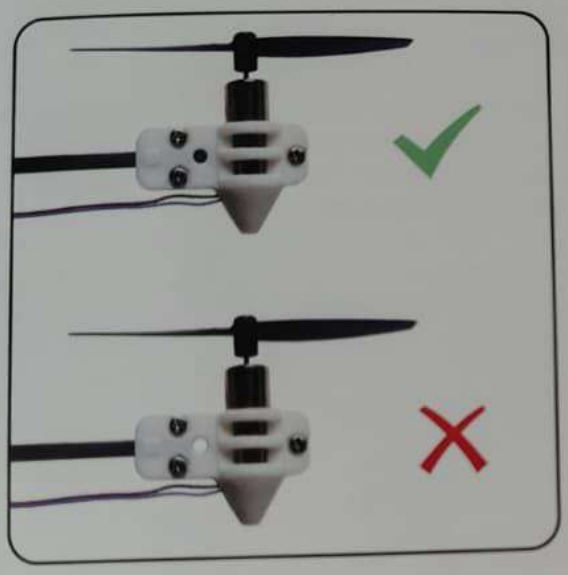
The F8620 features a 145 mm motor-to-motor diagonal distance, with propellers of 66 mm diameter .

2 MOTOR BRACKET

2





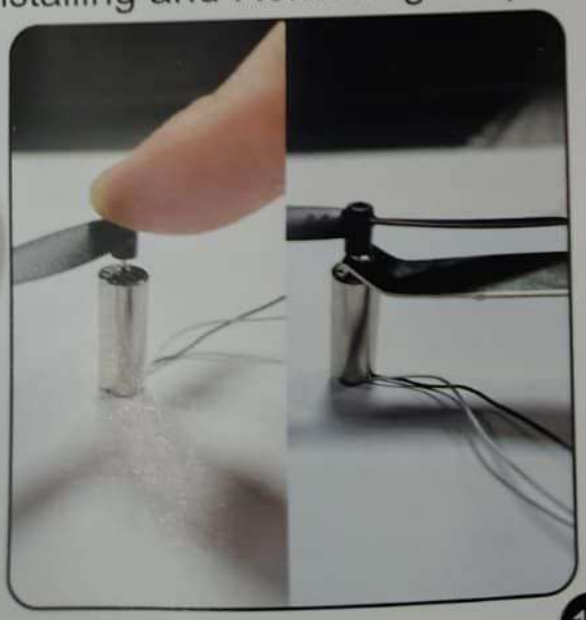


Tip:

The arrow on the frame shows the front.Install exactly as shown in the diagram above, matching the motor wire colors and propeller model numbers.

Tighten all screws one by one. Check through the pre-drilled hole to see if the carbon fiber rod is properly installed. The rod is correctly in place when the black carbon fiber rod fills the hole completely.

Installing and Removing Propellers



Install the propeller:

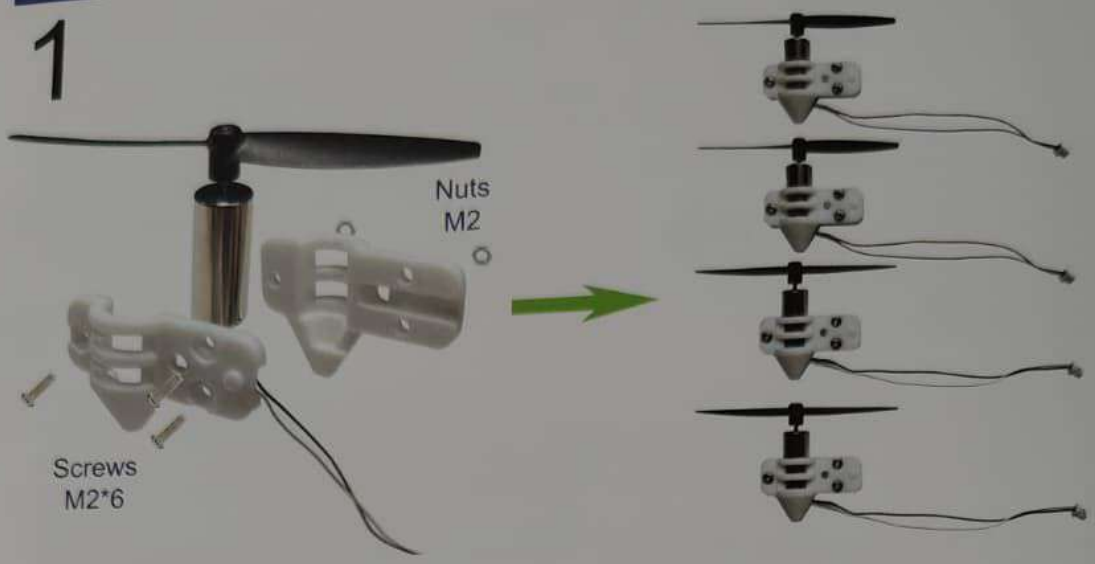
Carefully place the motor on a flat surface, then slowly press the propeller into place. The bottom of the coreless motor has a snap-fit design – pressing the propeller directly can damage the motor.

Removing the Propeller:

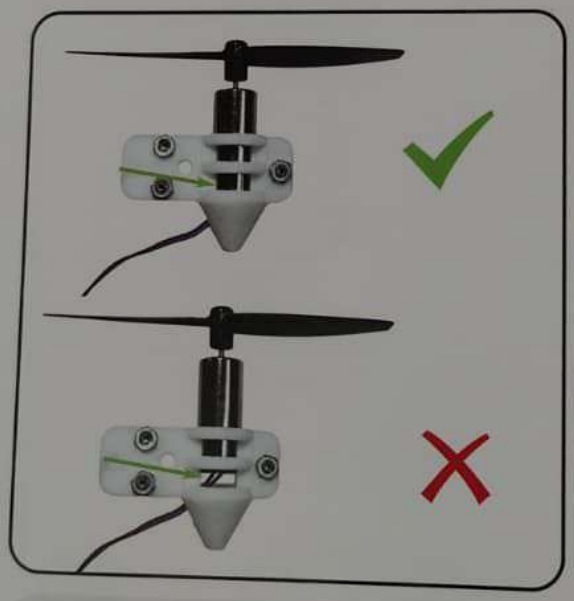
Use the tool from the accessory pack to slowly pry the propeller out.

2 MOTOR BRACKET

1



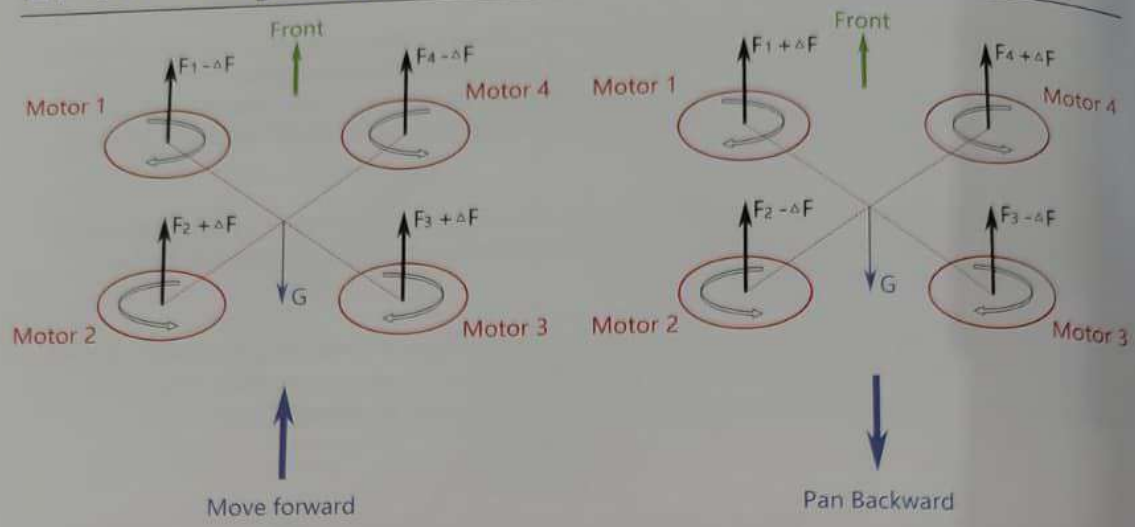
For each motor, use one motor mount marked with a circle and one marked with a square. Don't fully tighten the screws yet (leave them loose to adjust the carbon rods and motors).



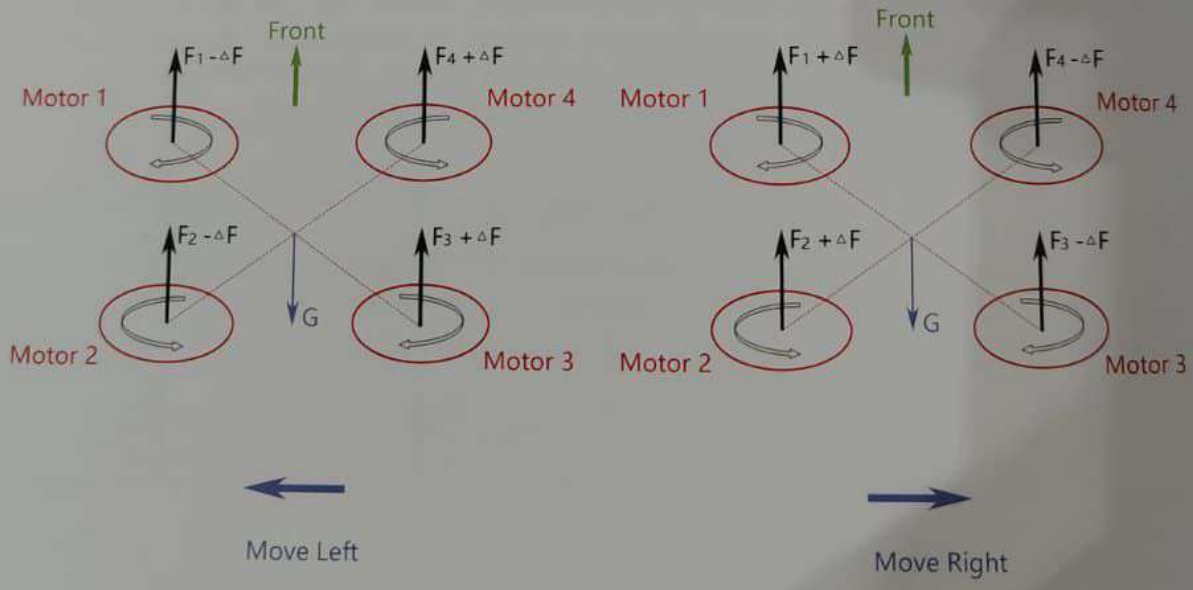
Tip:
When installing, make sure there are no gaps under the motor. Otherwise, the propeller won't stay level and will affect flight performance.

Tip:
Run the motor cables through the pre-drilled holes.

2. The Flying Principle



Similar to the principle of pitch motion, when the rotational speeds of the right rotors M3 and M4 increase while those of the left rotors M1 and M2 decrease, the lift generated by the right rotors M3 and M4 exceeds that of the left rotors M1 and M2, causing the quadrotor aircraft to move to the left. Conversely, if the rotational speeds of the left rotors increase while those of the right rotors decrease, it will move to the right. This is also known as roll motion.



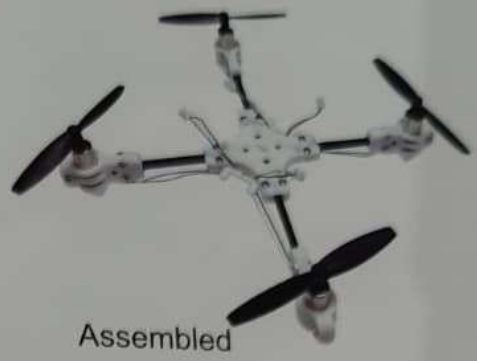
c. Yaw movement (rotating horizontally to the left, rotating horizontally to the right).

Ignoring external factors, when the rotation speeds of rotors M2 and M4 increase while the rotation speeds of rotors M1 and M3 decrease, the lift generated by M2 and M4 increases while the lift generated by M1 and M3 decreases. At this point, the total lift force F becomes zero, but the counterclockwise rotation speed is greater than the clockwise rotation speed, generating a counterclockwise torque that causes the drone to rotate to the left, also known as yawing to the left. Conversely, the drone will rotate to the right, also known as yawing to the right.

2 MOTOR BRACKET

3.2.1 Assemble the Motor Bracket

Pocket 2



Assembled

Components List

Coreless motors 8260
Red-blue wires – Clockwise 2x

Coreless motors 8260
Black-white wires – Counterclockwise 2x

Props – Clockwise (A) 2x

Props – Counterclockwise (B) 2x

Motor mounts
Circle mark 4x

Motor mounts
Square mark 4x

Screws
M2*6 12x

Nuts
M2 12x

1:1

The props come pre-installed.
Follow the diagram when replacing them to avoid damage.